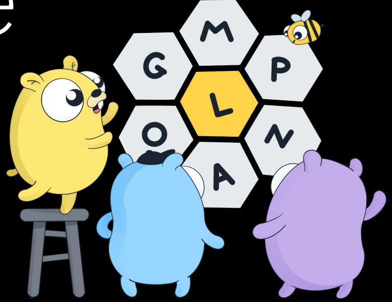
Supercharging ML Pipelines With Go

Leveraging Inter-Process Communication to streamline ML Inference using Go

The New York Times

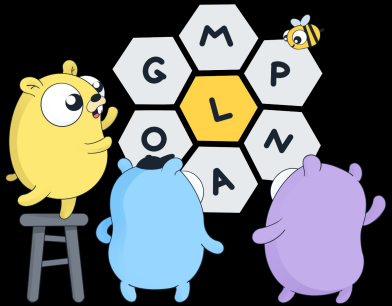


Design Credit: Rachel Shim | Renee French

Vaidehi Thete
Senior Software Engineer
Machine Learning & AI Team
Data Platform Mission

Machine Learning at Scale

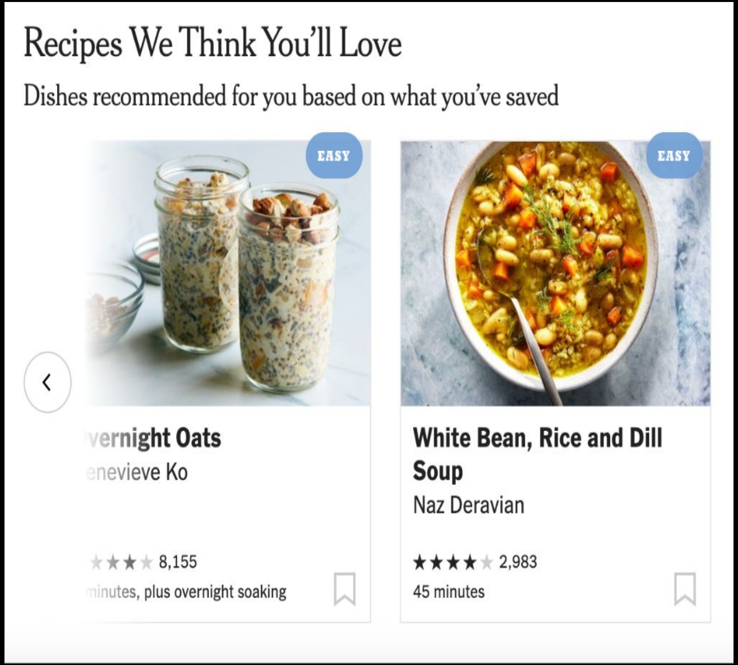
- **160+** machine learning models that power different products at the Times
- **4.5k+** requests per second
  at **<50ms p99**
- Operate multi-regionally in **central** and **east**

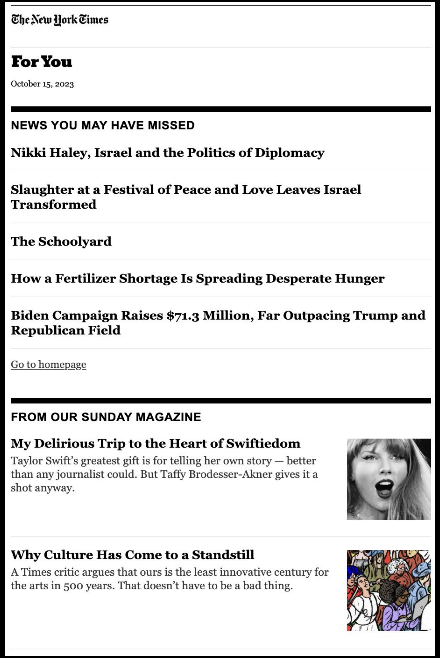


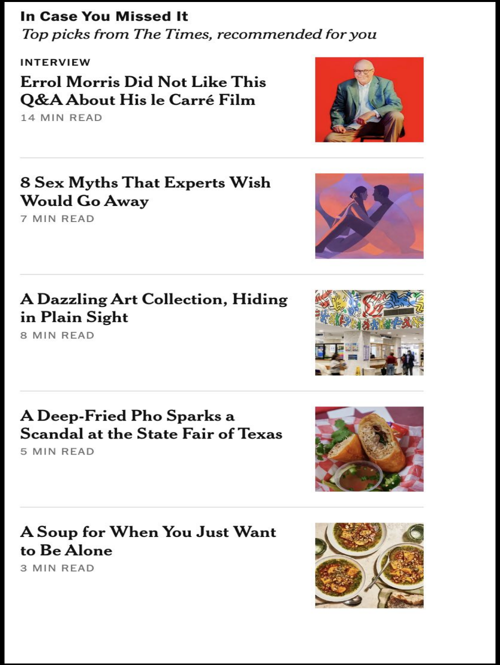
Our platform powers…

Where you may have seen us..

Engagement

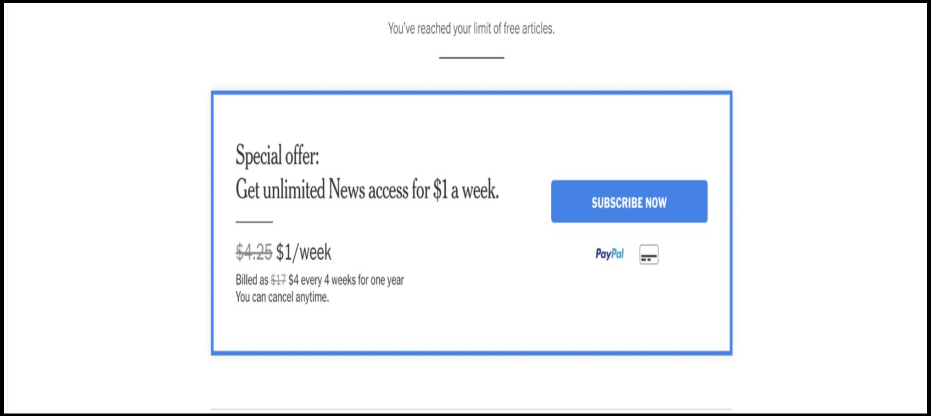


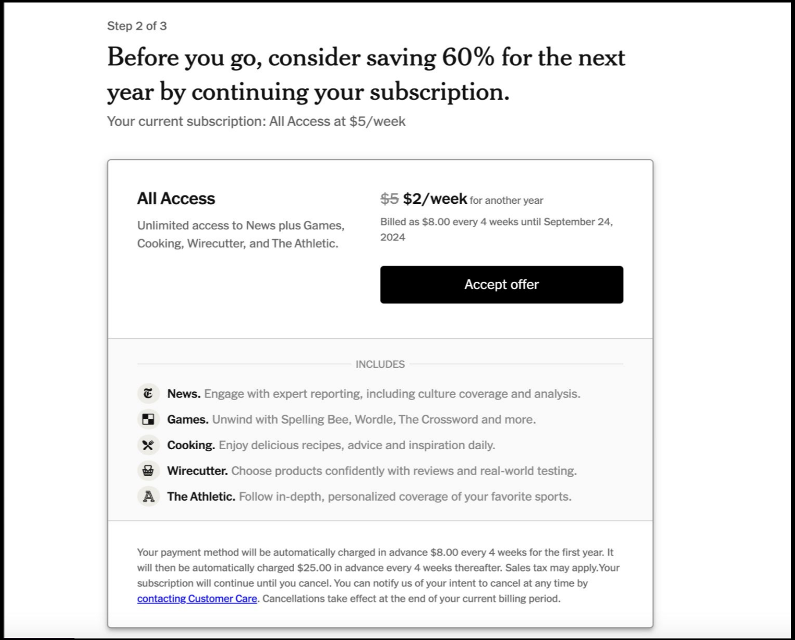




Where you may have seen us..

Growth

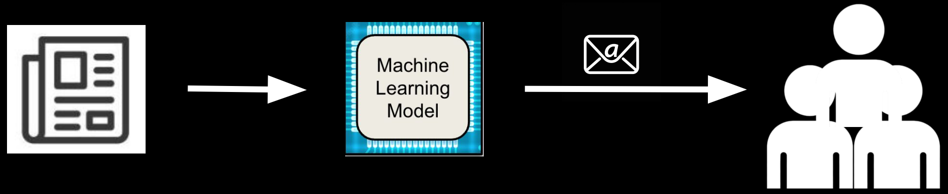




The New York Times

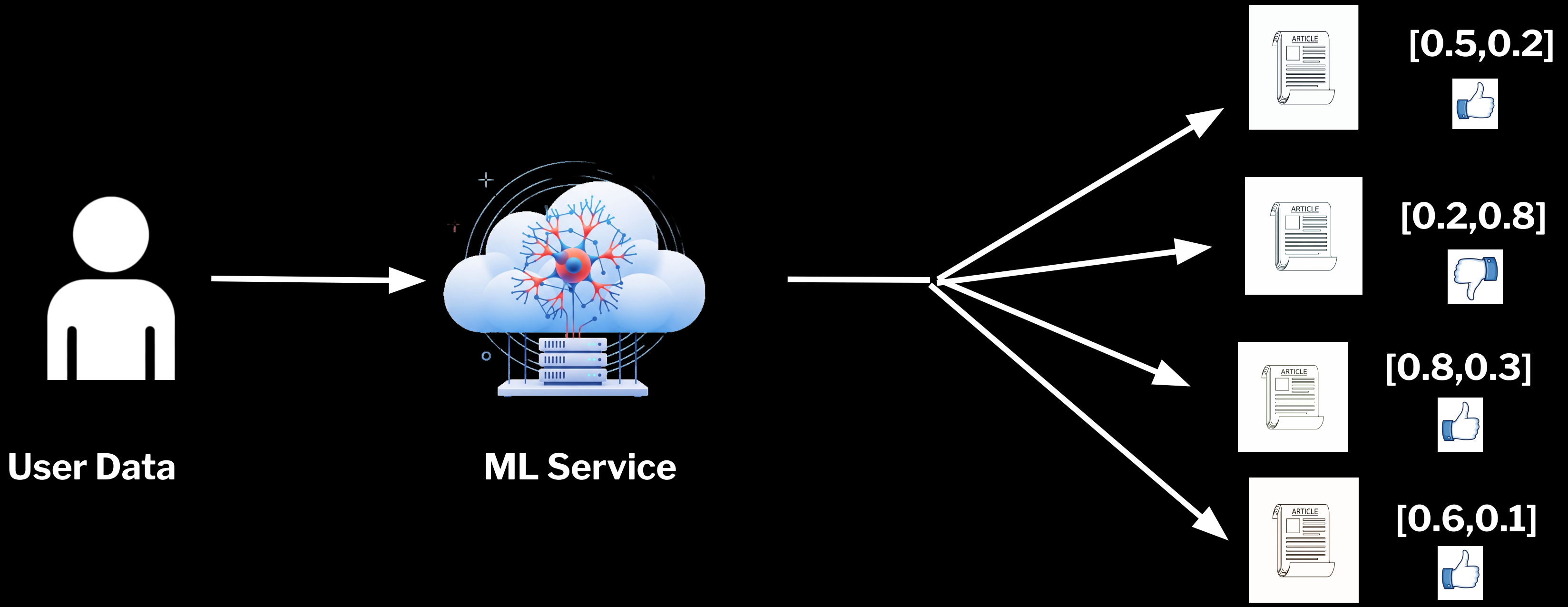
Targeted Emails

- Algorithmically identify a user base for a content
- By sending an email to the curated users, we drive the onsite traffic to the Times contributing to **+WAU** on average.



*A cute and excited animal deploying ML models to send emails to an **ML generated audience** of **Millions of Users***

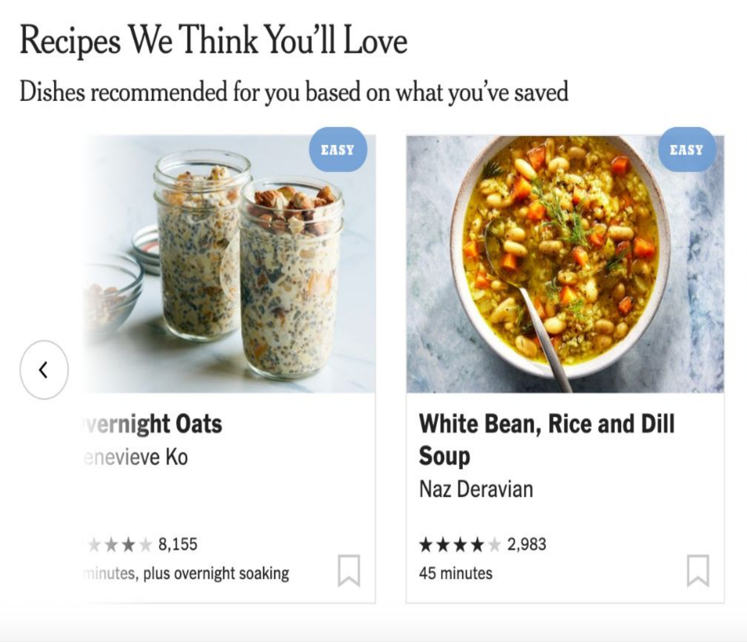
Part 1 : Single User Recommendations



The New York Times

Where you may have seen us.. (AGAIN)

Engagement



Recipes We Think You'll Love

Dishes recommended for you based on what you've saved

EASY

ernight Oats

enevieve Ko

8,155

minutes, plus overnight soaking

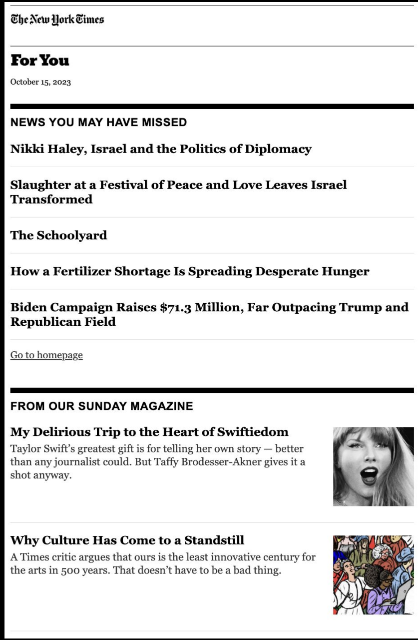
EASY

White Bean, Rice and Dill Soup

Naz Deravian

2,983

45 minutes



The New York Times

For You

October 15, 2023

NEWS YOU MAY HAVE MISSED

Nikki Haley, Israel and the Politics of Diplomacy

Slaughter at a Festival of Peace and Love Leaves Israel Transformed

The Schoolyard

How a Fertilizer Shortage Is Spreading Desperate Hunger

Biden Campaign Raises $71.3 Million, Far Outpacing Trump and Republican Field

Go to homepage

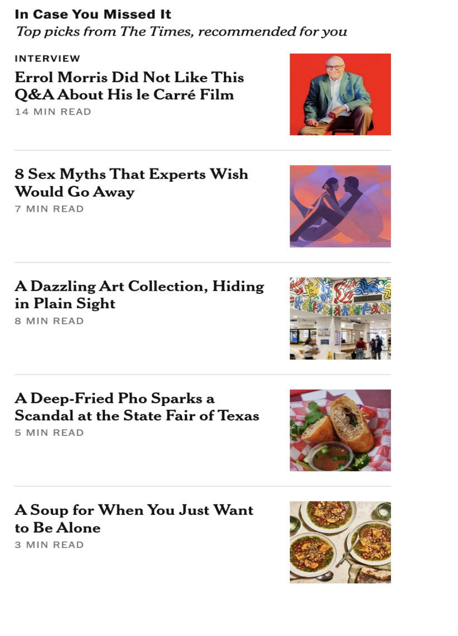
FROM OUR SUNDAY MAGAZINE

My Delirious Trip to the Heart of Swiftiedom

Taylor Swift's greatest gift is for telling her own story — better than any journalist could. But Taffy Brodesser-Akner gives it a shot anyway.

Why Culture Has Come to a Standstill

A Times critic argues that ours is the least innovative century for the arts in 500 years. That doesn't have to be a bad thing.



In Case You Missed It

Top picks from The Times, recommended for you

INTERVIEW

Errol Morris Did Not Like This Q&A About His le Carré Film

14 MIN READ

8 Sex Myths That Experts Wish Would Go Away

7 MIN READ

A Dazzling Art Collection, Hiding in Plain Sight

8 MIN READ

A Deep-Fried Pho Sparks a Scandal at the State Fair of Texas

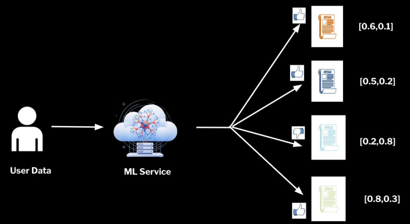
5 MIN READ

A Soup for When You Just Want to Be Alone

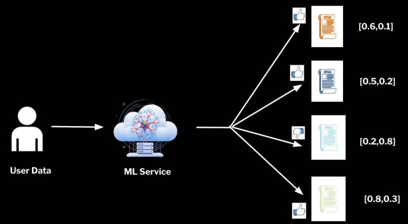
3 MIN READ

But this is something we already support. What's new?

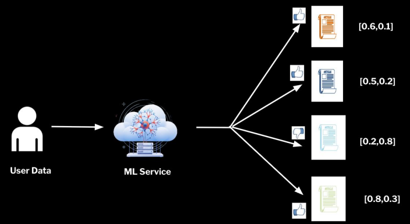
Part 2: Ranking 40M+ User-Content Recommendation Scores aggregated from a distributed service



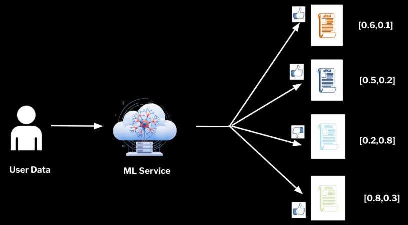
ML Service Replica 1



ML Service Replica 2

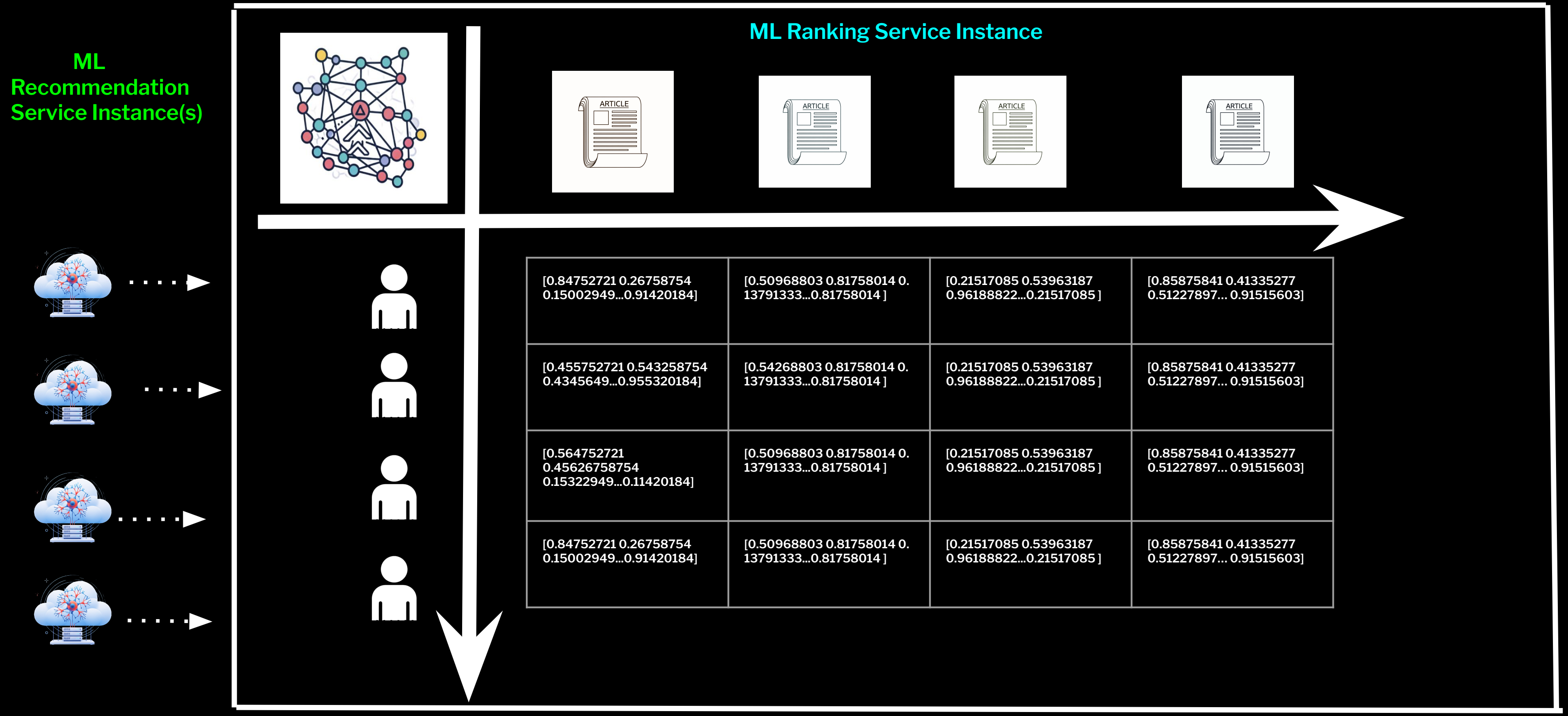


ML Service Replica 3

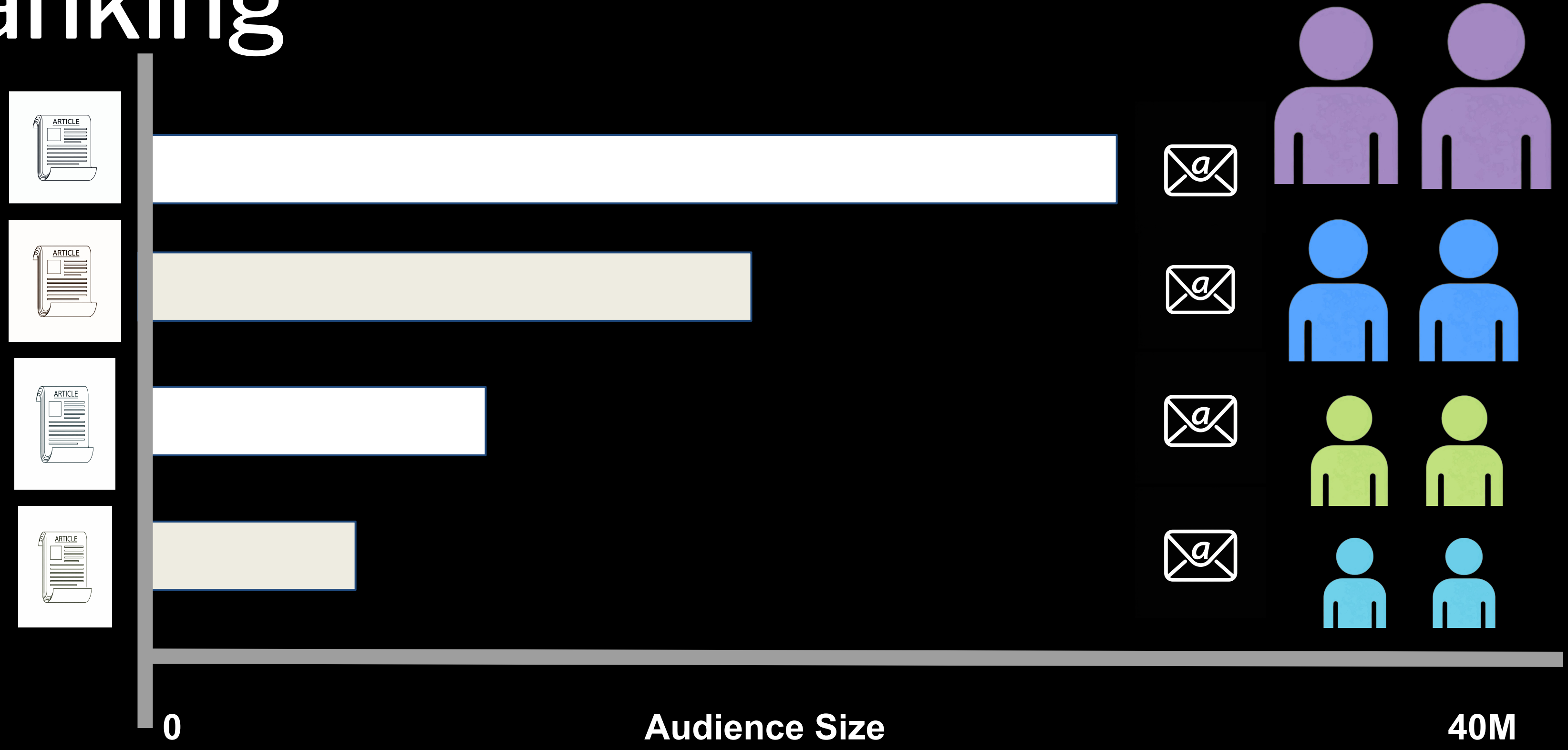


ML Service Replica N

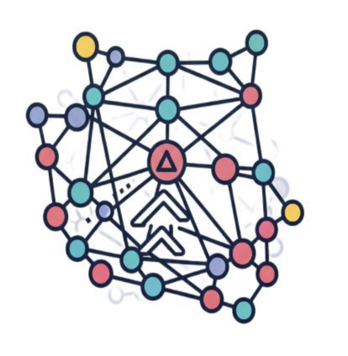
- **Independently score for 40M users**
- **Robust and Scalable**
- **Reduce latency**


ML Recommendation Service Instance(s)
ML Ranking Service Instance
ARTICLE
ARTICLE
ARTICLE
ARTICLE
[0.84752721 0.26758754 0.15002949...0.91420184]
[0.50968803 0.81758014 0. 13791333...0.81758014]
[0.21517085 0.53963187 0.96188822...0.21517085]
[0.85875841 0.41335277 0.51227897... 0.91515603]
[0.455752721 0.543258754 0.4345649...0.955320184]
[0.54268803 0.81758014 0. 13791333...0.81758014]
[0.21517085 0.53963187 0.96188822...0.21517085]
[0.85875841 0.41335277 0.51227897... 0.91515603]
[0.564752721 0.45626758754 0.15322949...0.11420184]
[0.50968803 0.81758014 0. 13791333...0.81758014]
[0.21517085 0.53963187 0.96188822...0.21517085]
[0.85875841 0.41335277 0.51227897... 0.91515603]
[0.84752721 0.26758754 0.15002949...0.91420184]
[0.50968803 0.81758014 0. 13791333...0.81758014]
[0.21517085 0.53963187 0.96188822...0.21517085]
[0.85875841 0.41335277 0.51227897... 0.91515603]

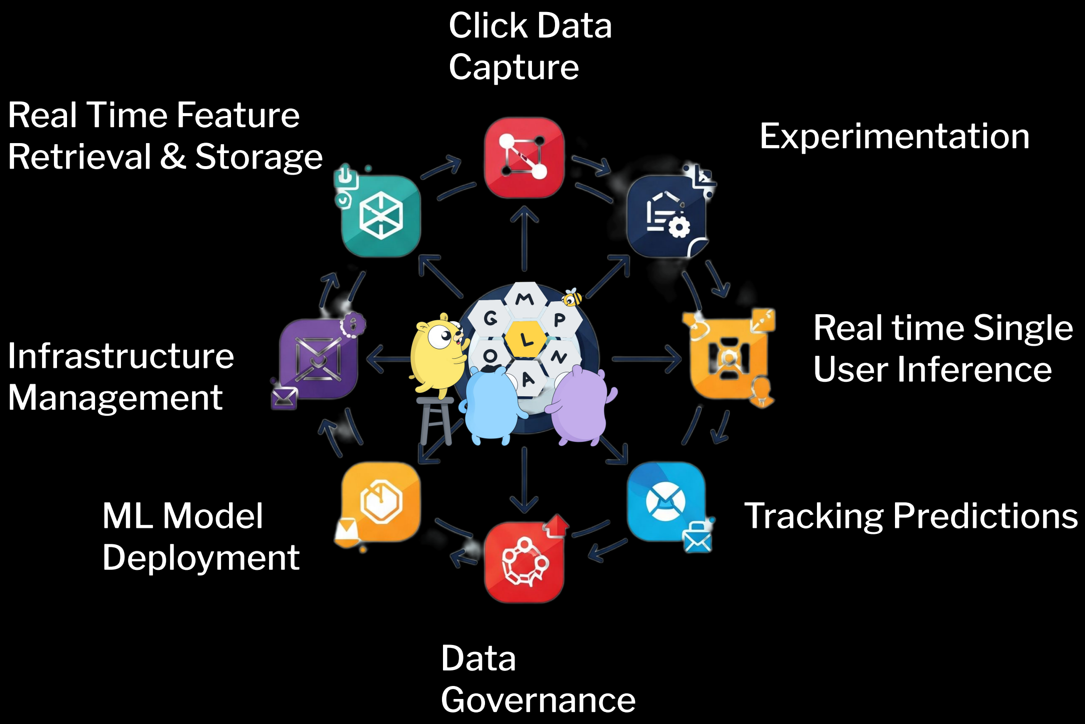
Ranking



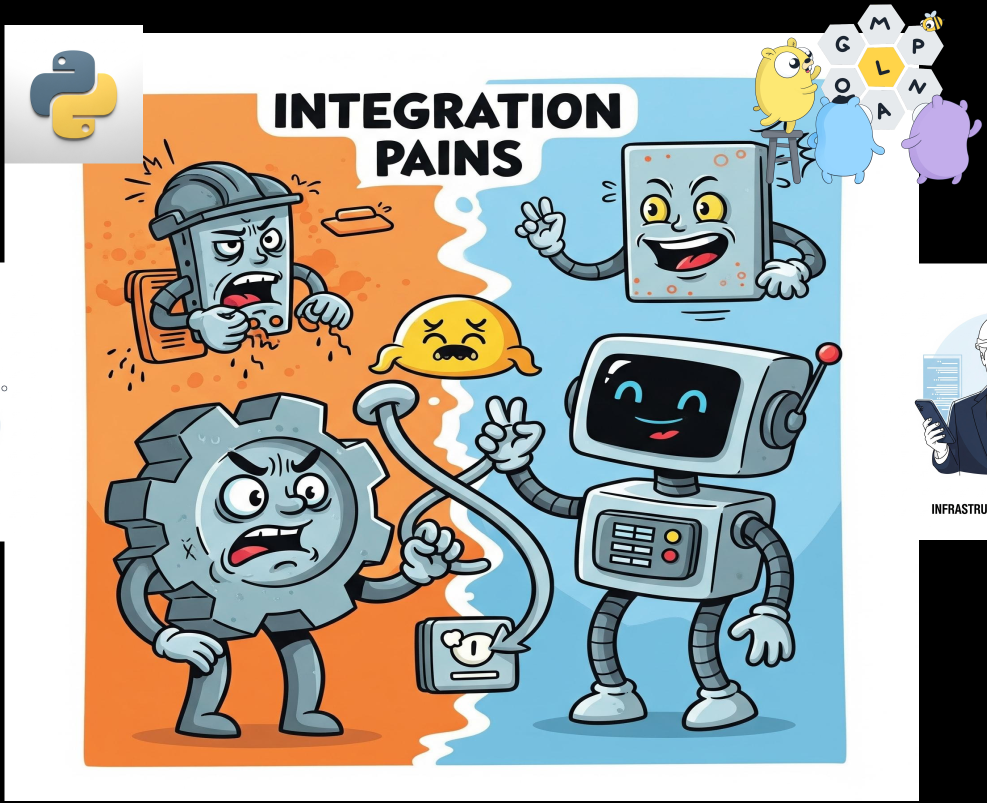
The New York Times

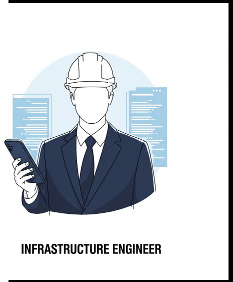


Ranking Model for 40M+ data points



Problem Statement

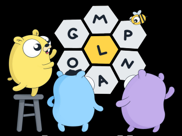




Separation of Concerns

Ranking Algorithm

- Logic owned by DS
- Business Logic
- Data Munging



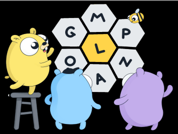
Operationalization

- Owned by ML & AI Platform Engineers
- Data Transfer
- State Management
- Metadata Addition
- Type Safety
- Ingress and Egress

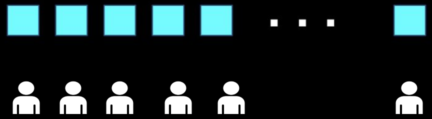
Minimal Data Transfer Overhead & Safe Writes

python

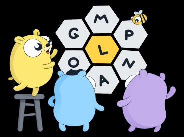
- Dimension Checks for Scores



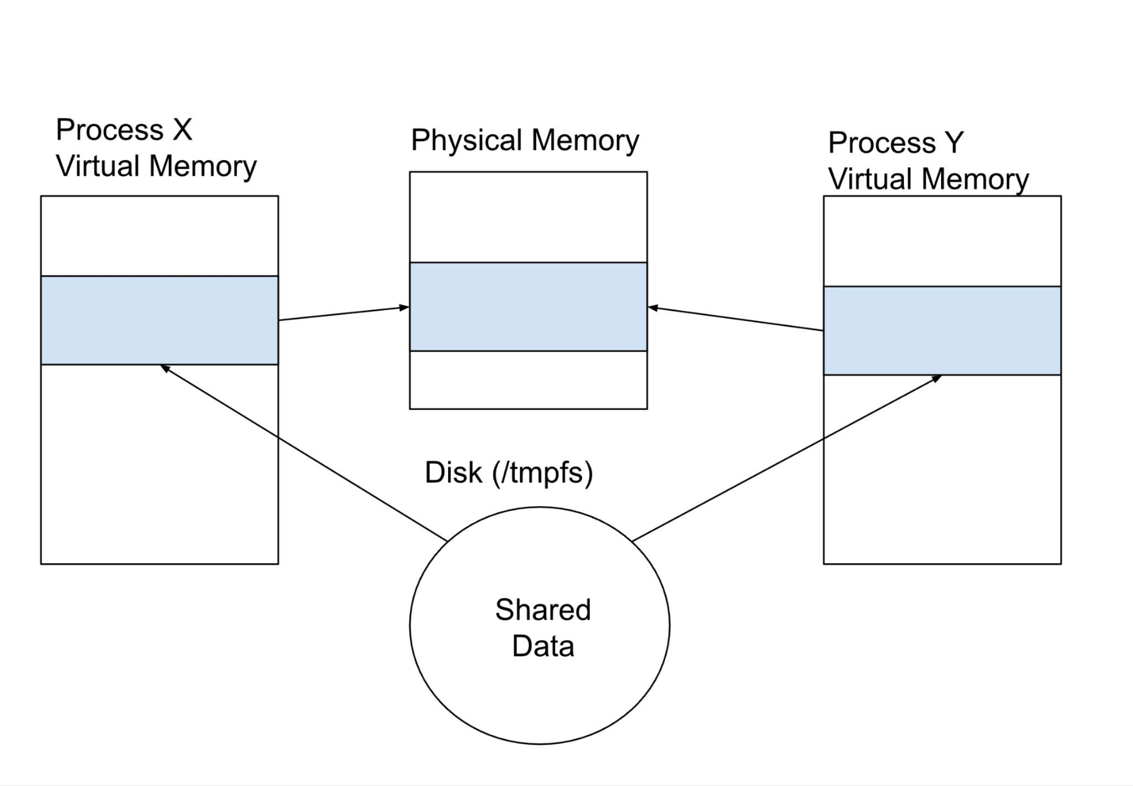
- Batching of Scores
- Data Integrity of Scores

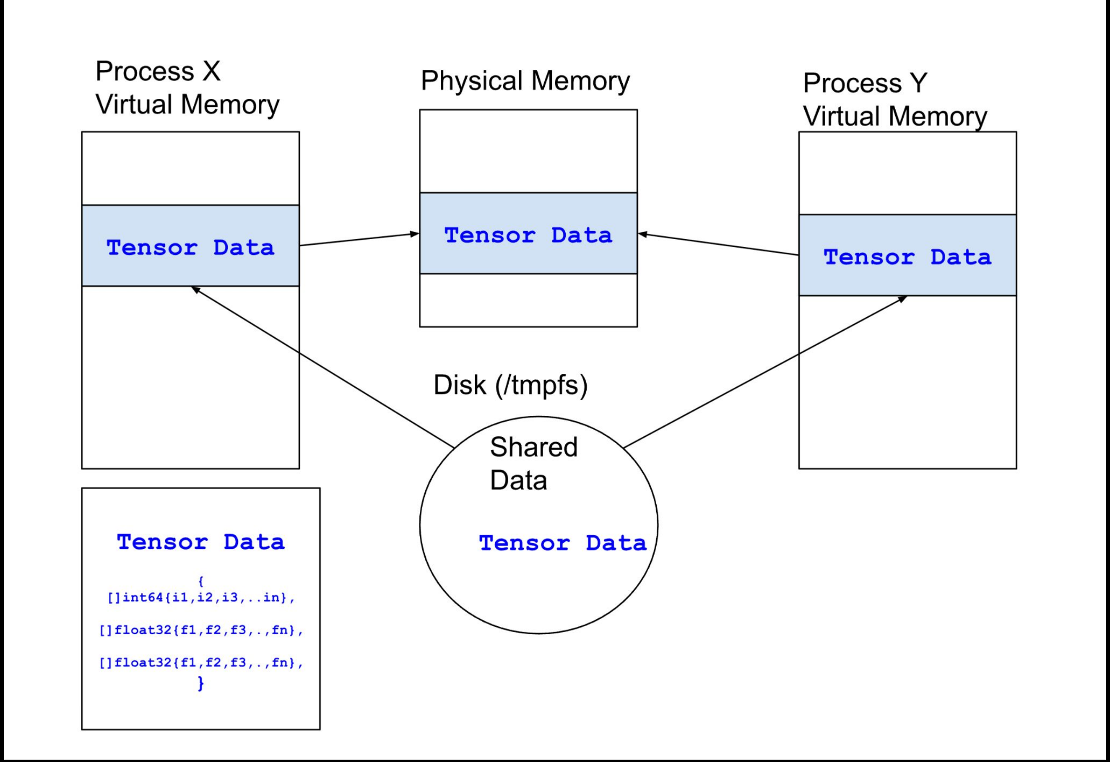


Robust to Failures: Idempotent

Stateless Entity	 Restart the operations anew Maintain fast reads and data integrity

Shared Memory


Process X
Virtual Memory
Physical Memory
Process Y
Virtual Memory
Disk (/tmpfs)
Shared
Data


Process X
Virtual Memory
Tensor Data
Physical Memory
Tensor Data
Process Y
Virtual Memory
Tensor Data
Disk (/tmpfs)
Shared
Data
Tensor Data
Tensor Data
{
[]int64{i1,i2,i3,..in},
[]float32{f1,f2,f3,.,fn},
[]float32{f1,f2,f3,.,fn},
}

Shm Creation : Initialization

Operationalization

```
import "syscall"
func createSHMRegion(name string, size int64) (int,[]byte, error){
      // create the file object
      fd, err := syscall.Open(shmName,syscall.O_CREAT|syscall.O_RWR,0666)
      if err != nil{
            return -1, fmt.Errorf("failed to create shm %v, %v", name, err)

      }

     //set the size of the shm memory
      err = syscall.Ftruncate(fd, int64(size)
     if err != nil{
            return -1, fmt.Errorf("failed to set shm size for %v, %v", name, err)

      }


      // map the file into the memory region
      data, err := syscall.Mmap(fd, 0, size,syscall.PROT_READ|syscall.PROT_WRITE, syscall.MAP_SHARED)
      if err != nil {
            return -1, nil, fmt.Errorf("failed to map shared memory: %w", err)
      }

     return fd, data,nil

}
```

Shm Creation : Copy Data to SHM

Operationalization

```go
func copyToSHM(shmData []byte, offset int,
tensorData[]byte) (int, error) {

    // length check
    if offset+len(newData) > len(data) {
        return offset, fmt.Errorf("not enough
        space in shared memory")
    }

    // copy bytes to shm
    copy(shmData[offset:], tensorData)
    return offset + len(newData), nil
}
```

Shm Creation : Delete SHM Region

Operationalization

```go
func cleanupSharedMemory(name string, data []byte, fd int) error {
    // Unmap memory from current process
    if err := syscall.Munmap(data); err != nil {
        return fmt.Errorf("munmap failed: %w", err)
    }

    // Close the file descriptor
    if err := syscall.Close(fd); err != nil {
        return fmt.Errorf("close failed: %w", err)
    }

    // Remove the shared memory file
    if err := syscall.Unlink(name); err != nil {
        return fmt.Errorf("unlink failed: %w", err)
    }

    return nil
}
```

Shm Creation : Read from SHM

Operationalization

```
func readFromSHM(shm []byte) ([]int64,error) {

 result := make([]int64,0)

 numElements := len(shm) / 8

 // read a shm containing a slice of int64 integers
 for i := 0; i < numElements; i++ {

        val := int64(binary.LittleEndian.Uint64(shm[i*8:i*8+8]))
        result = append(result, val)
    }

 return result,nil
}
```

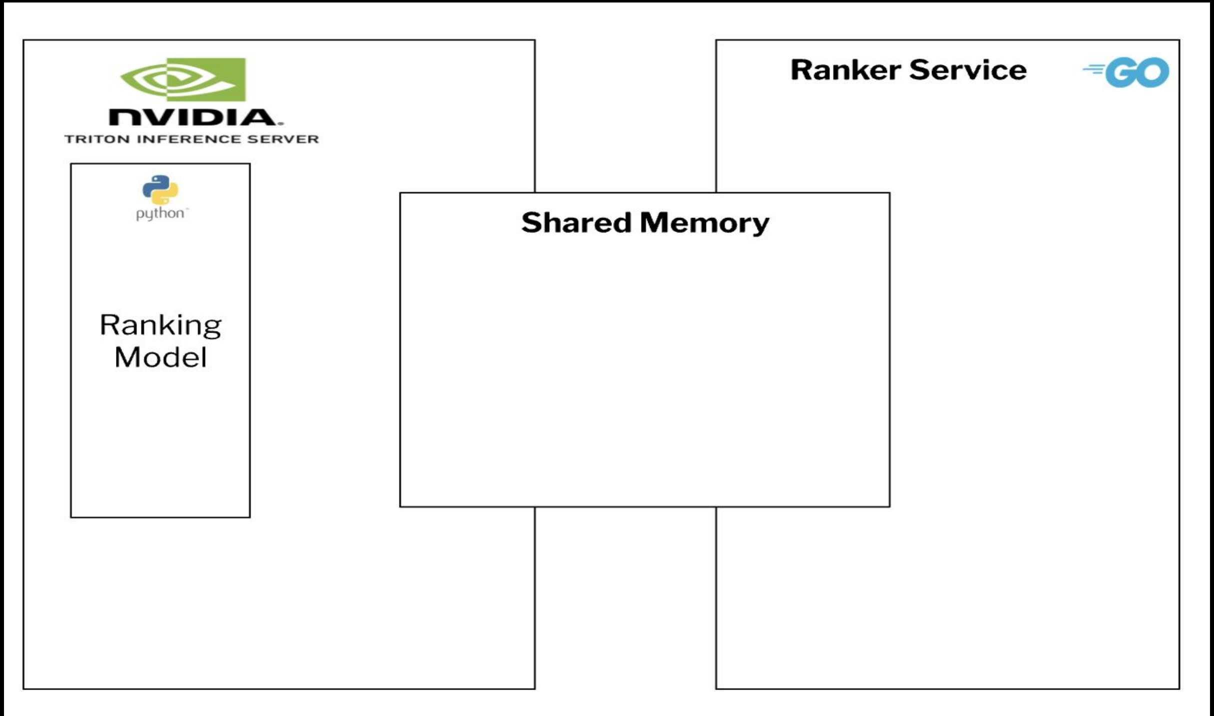
Core Operations for SHM Management

Operationalization

```go
func createSHMRegion(name string, size int64) (int,[]byte, error)

func copyToSHM(shmData []byte, offset int, tensorData[]byte) (int, error)

func cleanupSharedMemory(name string, data []byte, fd int) error

func readFromSHM(shm []byte) (*Data,error) //Data= your own data
```

Separation of Concerns

Inter Process Communication via Shared Memory



Word of the Day: Tensor

A **tensor** is a fundamental data structure in machine learning that represents a multidimensional array of numerical values. Think of it as a container for data with different dimensions:

0D Tensor (Scalar)

```
scalar = 5.0
```

1D Tensor (Vector)

```
vector = [1.0, 2.0,
3.0, 4.0]
```

2D Tensor (Matrix)

```
matrix = [
    [1.0, 2.0, 3.0],
    [4.0, 5.0, 6.0]
]
```

3D Tensor

```
image_tensor = [
    [[255, 0, 0], [0, 255, 0]],  #
Red and Green pixels
    [[0, 0, 255], [255, 255, 0]]  #
Blue and Yellow pixels
]
```

… and so on

Single Instance Score

```go
type (

	Scores struct{

		Tensor1 int64
		Tensor2 [][]float32
		Tensor3 []int8
		Tensor4 []float32

	}

)
```

Scores = Output of the ML Recommendation Service per user

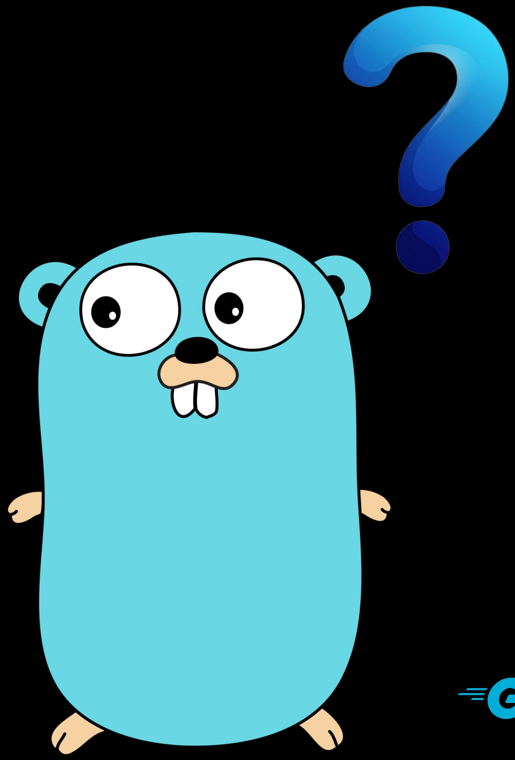
Size calculation:

Tensor Name	**Tensor Type**	**Number of entities per Record**	**Total Size of Tensor**
Tensor 1	int64	1	8 bytes
Tensor 2	[][]float32	5 * 2	40 bytes
Tensor 3	int8	5	5 bytes
Tensor 4	[]float32	1	4 bytes

Total Size per record: 57 bytes

Minimal Data Transfer Overhead & Safe Writes

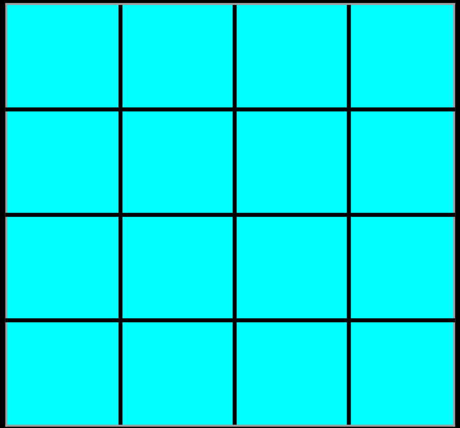
How do we ensure writes of 40M records each occupying about ~60 bytes of storage in a way that is safe and fast?



Minimal Data Transfer Overhead & Safe Writes

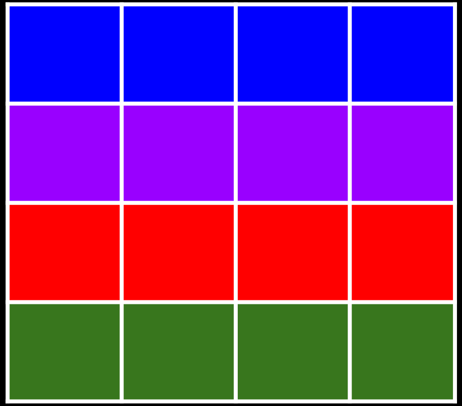
Convert 500 individual scores into a batch of tensor columns

A batch of 500 records



= 31 individual scores,
4 surplus in the last batch

The same batch of 500 scores grouped by tensors



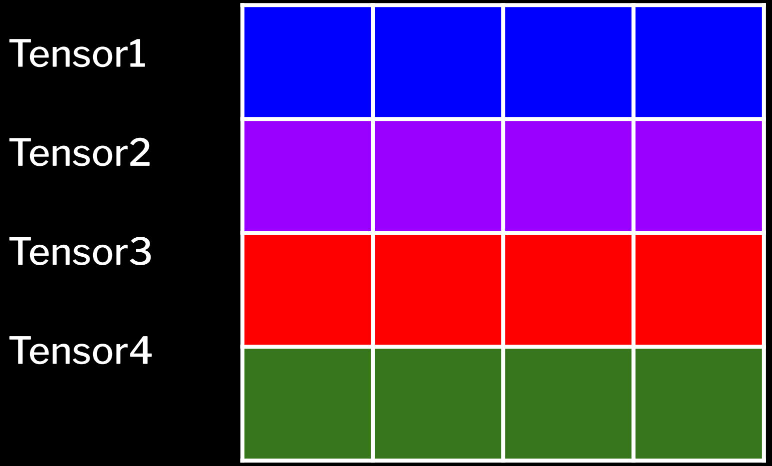
per record	500 records size
1	500
5 * 2	5000
5	2500
1	500

```go
type (

    Scores struct{

        Tensor1 int64
        Tensor2 [][]float32
        Tensor3 []int8
        Tensor4 []float32

    }

)
```

Minimal Data Transfer Overhead & Safe Writes

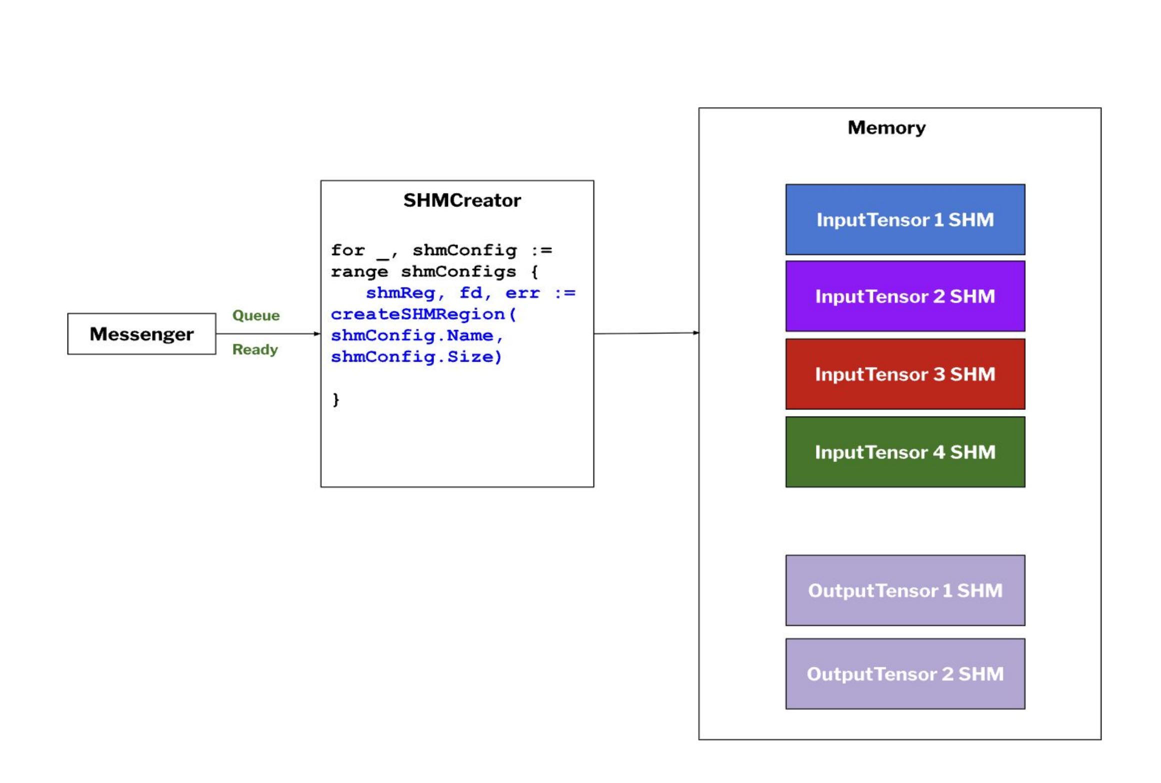
Goal : Batching to speed up data transfer

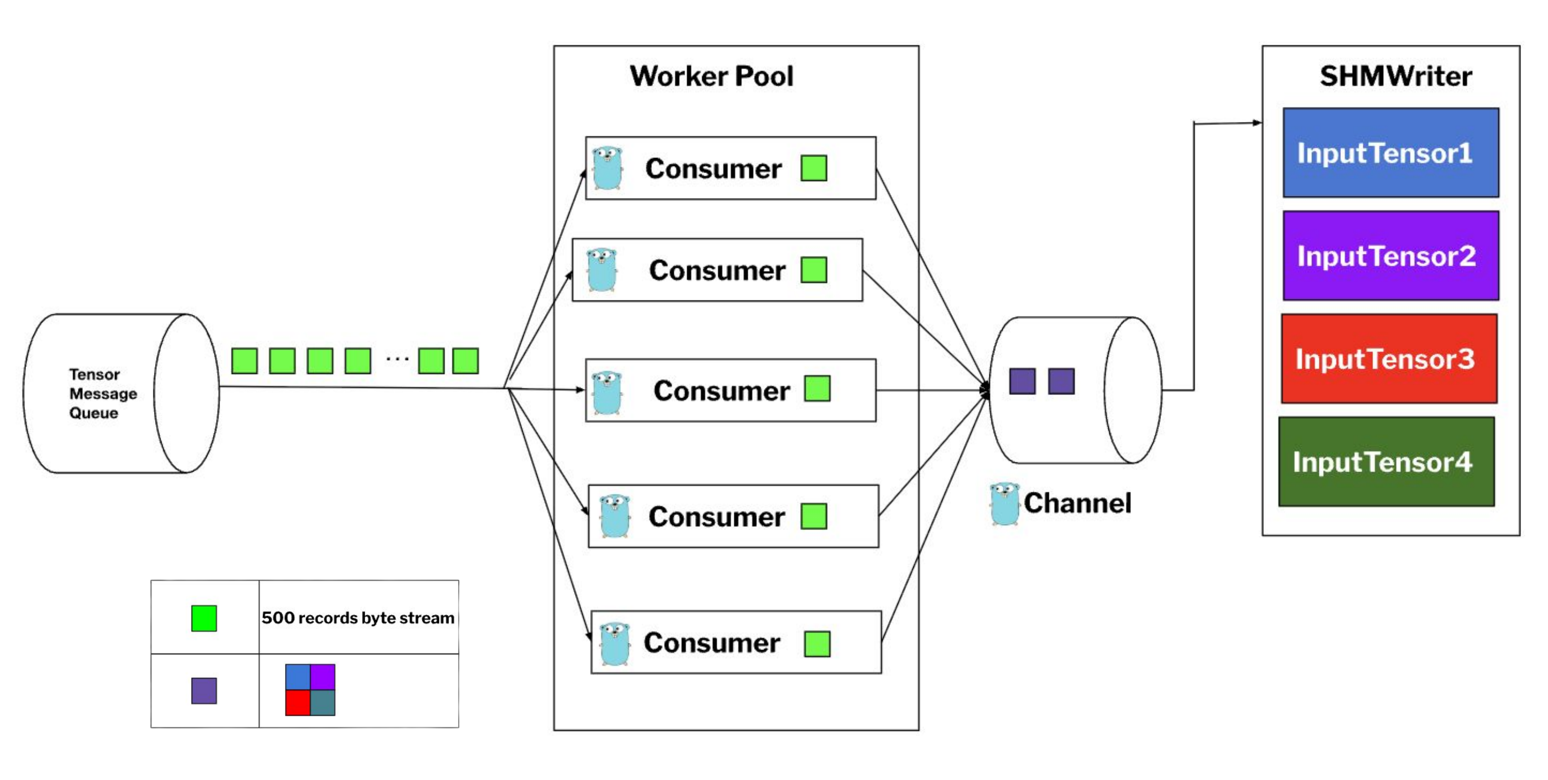
```go
type (

    DataPack struct{

        Tensor1 []byte

        // [][]float32 2D array
        // converted to row major order
        Tensor2 []byte

        Tensor3 []byte

        Tensor4 []byte

        Checksum map[string][16]byte

    }

)
```



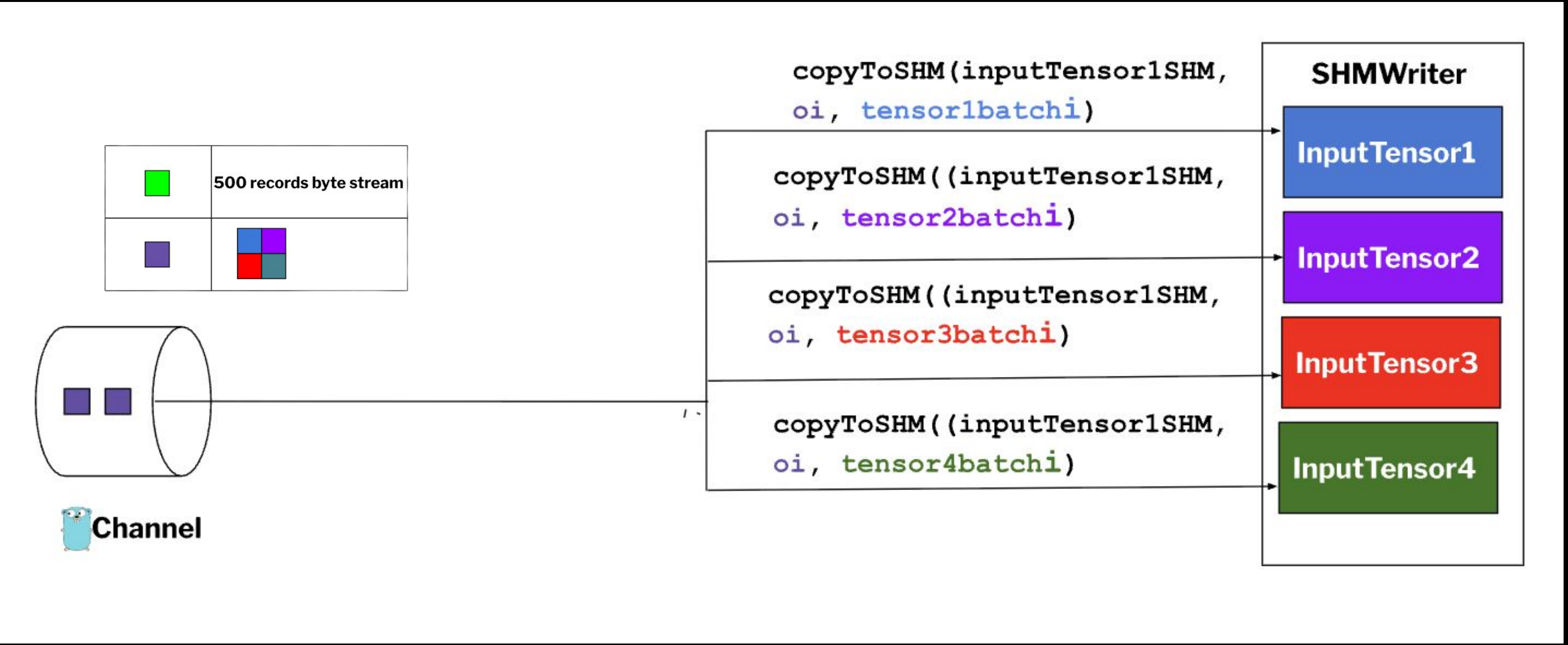
Checksum for each batch is used to validate the integrity of the data before it is passed off for inference

SHM Region: Creation




Worker Pool
Consumer
Consumer
Consumer
Consumer
Consumer
SHMWriter
InputTensor1
InputTensor2
InputTensor3
InputTensor4
Tensor Message Queue
Channel
500 records byte stream

SHM Population : Ingestion and Processing
Copy Operation

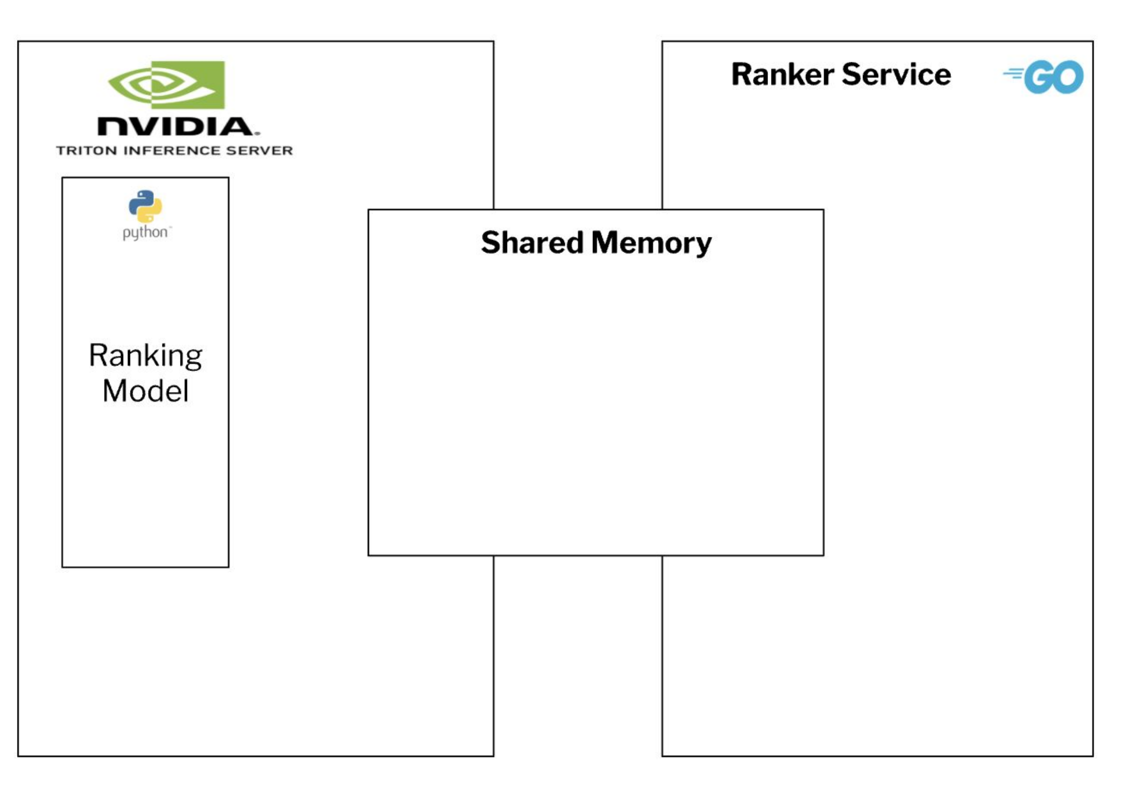


Idempotent to failures and faster restart

On average it takes about **50s** to write **40M** records(**~4GB**) to SHMs making it fairly robust to failures and leading to faster retries for the whole dataset

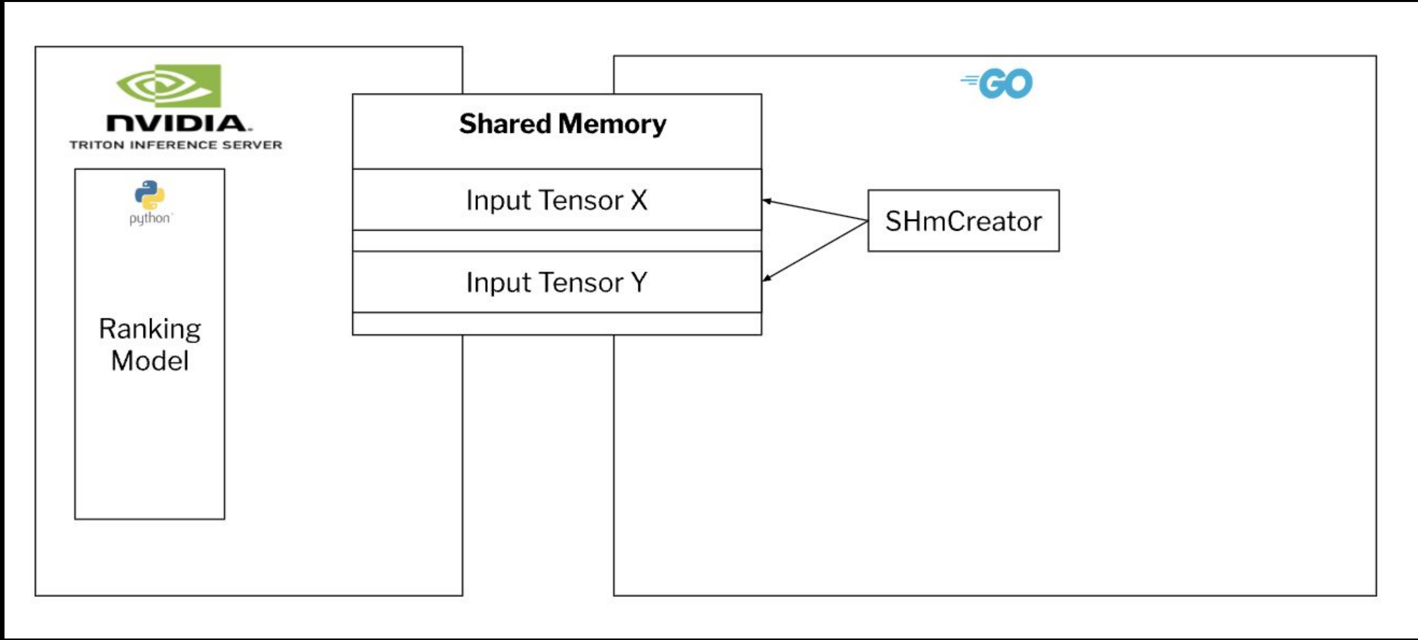
How it all comes together…

Step 1 : Startup the Processes



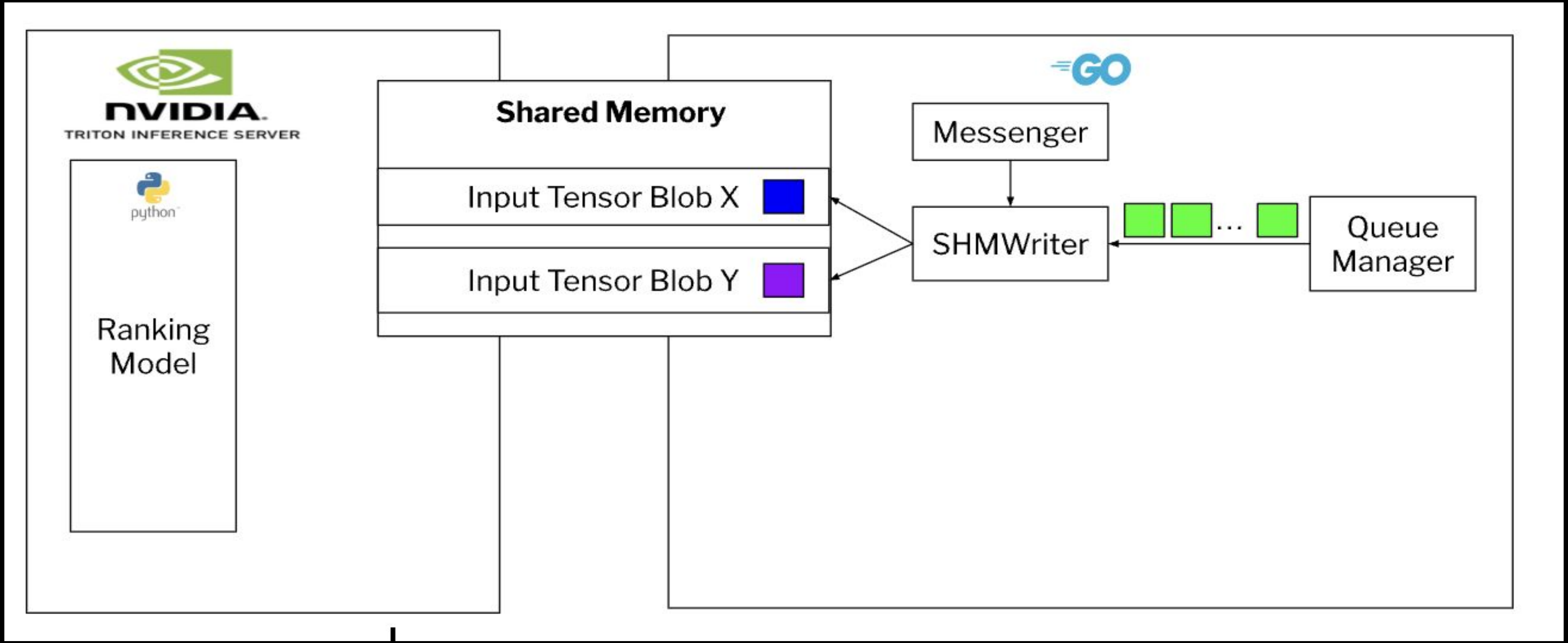
How it all comes together…

Step 2 : Create the SHMs per tensor

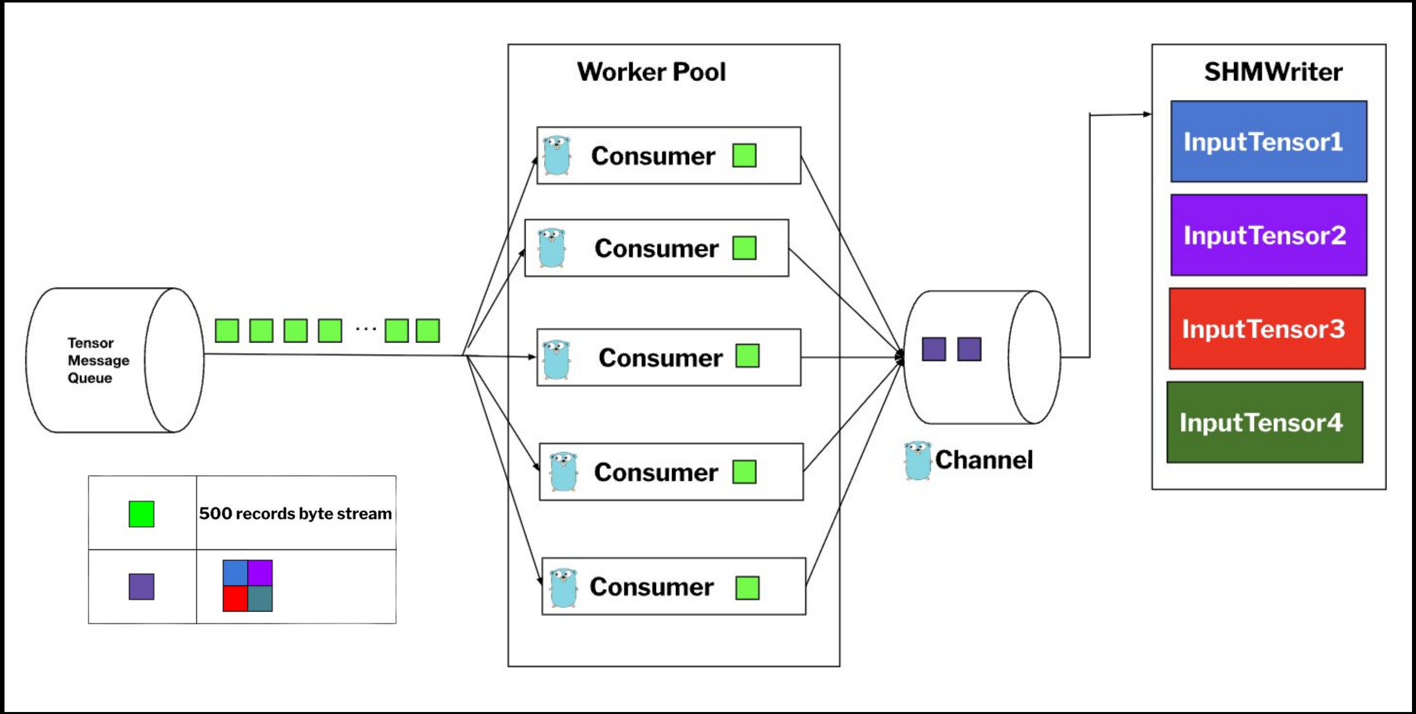


How it all comes together...

Step 3: Populate SHM for each tensor
upon queue having data

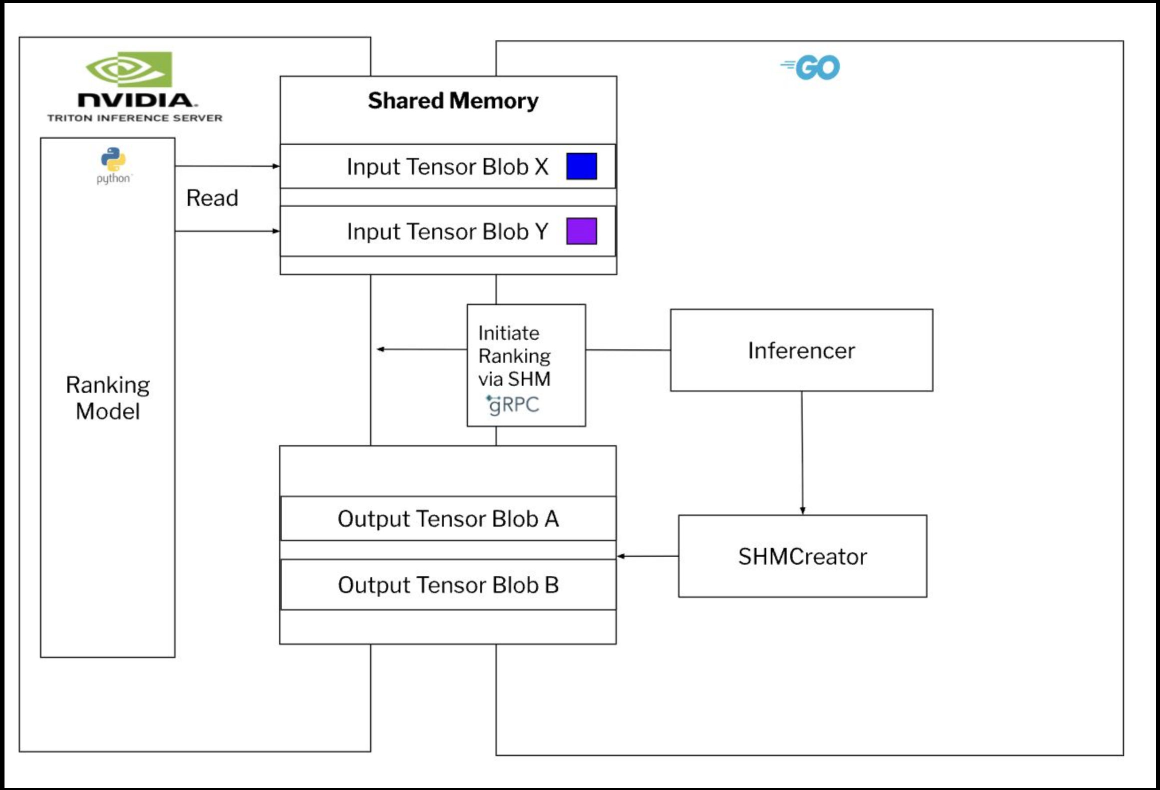


The New York Times


Worker Pool
SHMWriter
Consumer
Consumer
Consumer
Consumer
Consumer
Tensor Message Queue
Channel
InputTensor1
InputTensor2
InputTensor3
InputTensor4
500 records byte stream

How it all comes together…

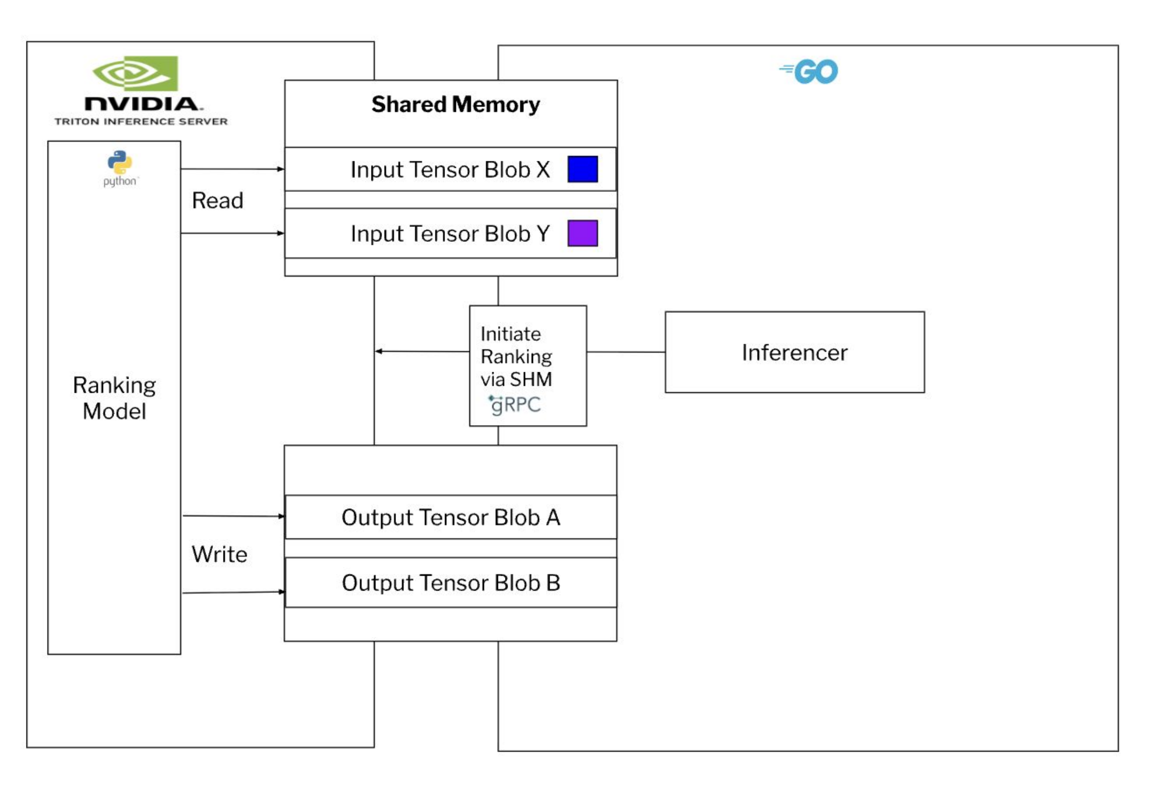
Step 4: Hand off tensors to Ranking Model
to Python Service



The New York Times

How it all comes together…

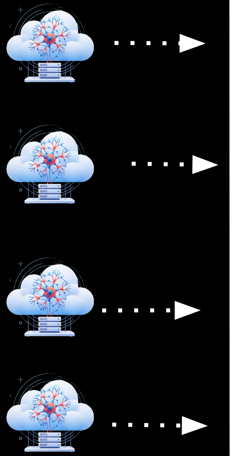
Step 5: Perform Ranking



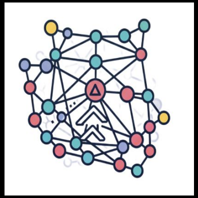
The New York Times

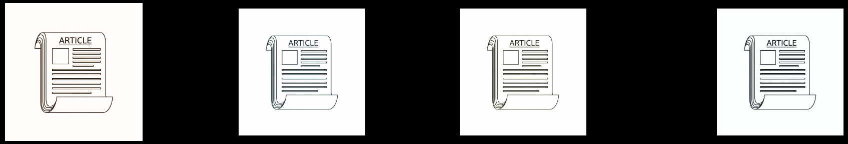
Wait! What is Ranking again?

ML Recommendation Service Instance(s)



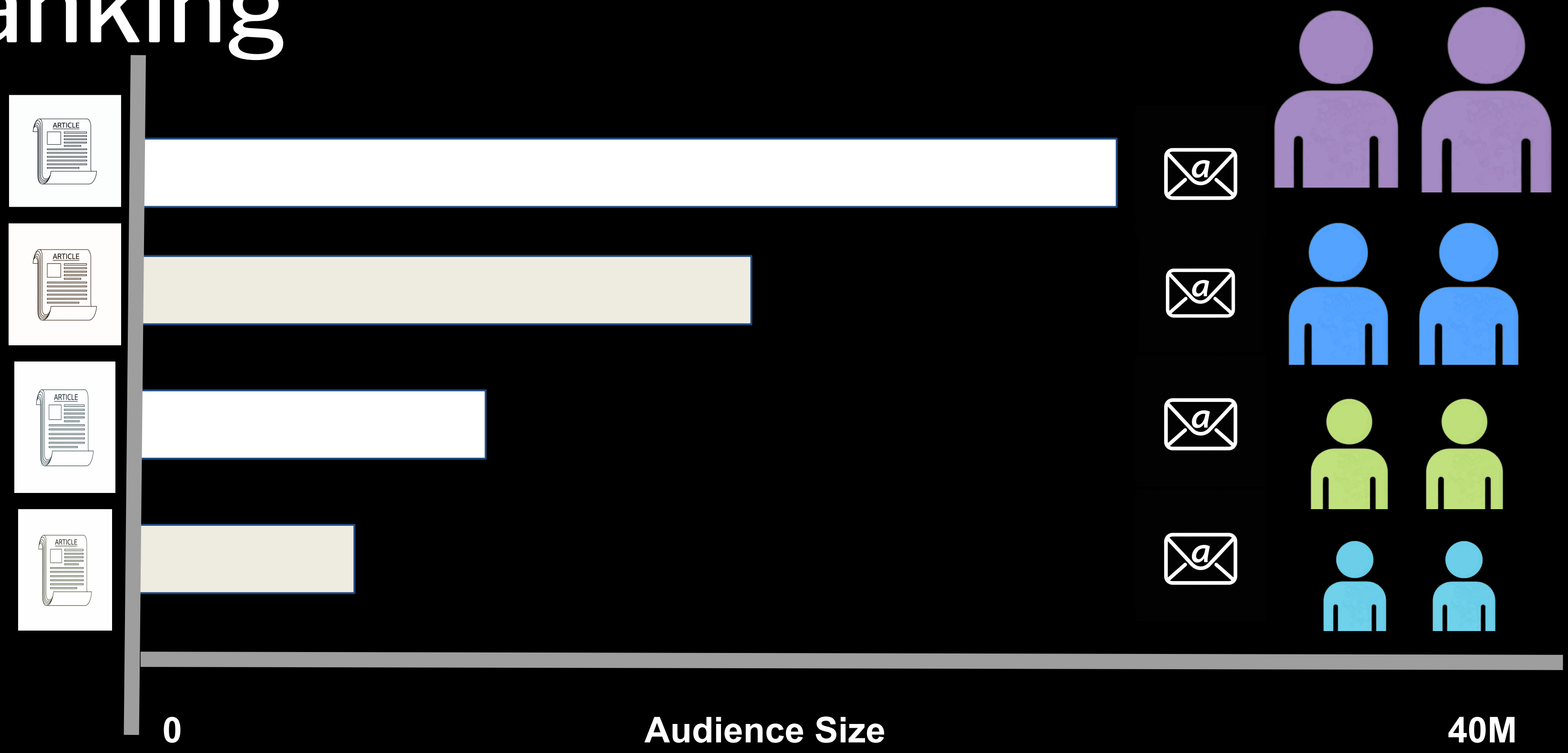
ML Ranking Service Instance





[0.84752721 0.26758754 0.15002949...0.91420184]	[0.50968803 0.81758014 0. 13791333...0.81758014]	[0.21517085 0.53963187 0.96188822...0.21517085]	[0.85875841 0.41335277 0.51227897... 0.91515603]
[0.455752721 0.543258754 0.4345649...0.955320184]	[0.54268803 0.81758014 0. 13791333...0.81758014]	[0.21517085 0.53963187 0.96188822...0.21517085]	[0.85875841 0.41335277 0.51227897... 0.91515603]
[0.564752721 0.45626758754 0.15322949...0.11420184]	[0.50968803 0.81758014 0. 13791333...0.81758014]	[0.21517085 0.53963187 0.96188822...0.21517085]	[0.85875841 0.41335277 0.51227897... 0.91515603]
[0.84752721 0.26758754 0.15002949...0.91420184]	[0.50968803 0.81758014 0. 13791333...0.81758014]	[0.21517085 0.53963187 0.96188822...0.21517085]	[0.85875841 0.41335277 0.51227897... 0.91515603]

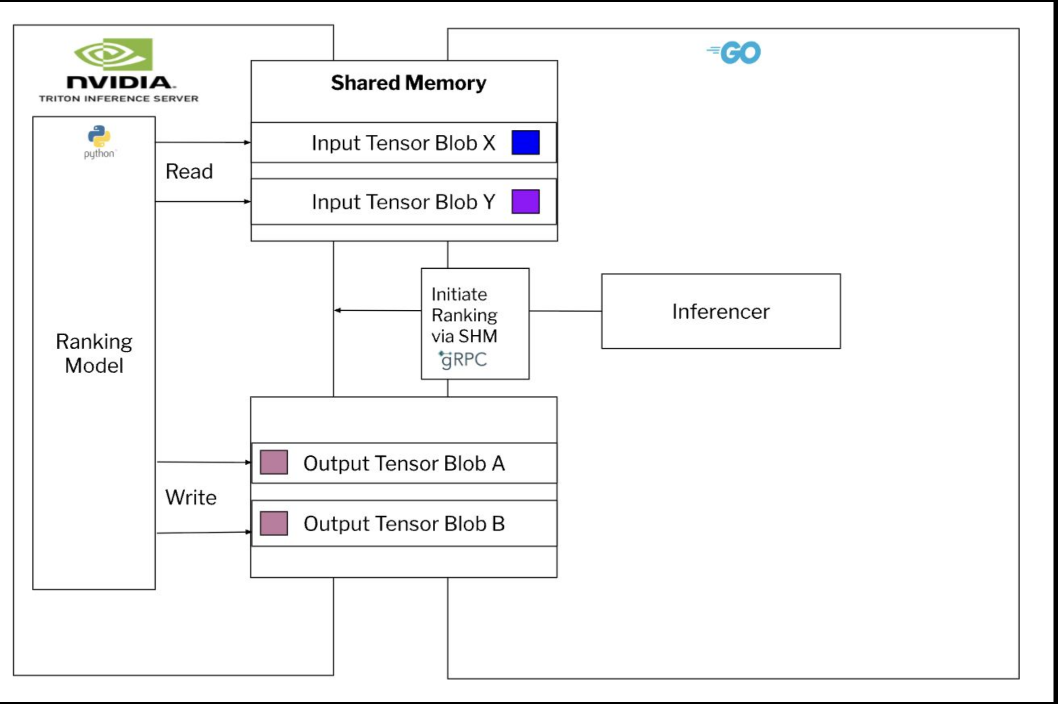
Ranking



The New York Times

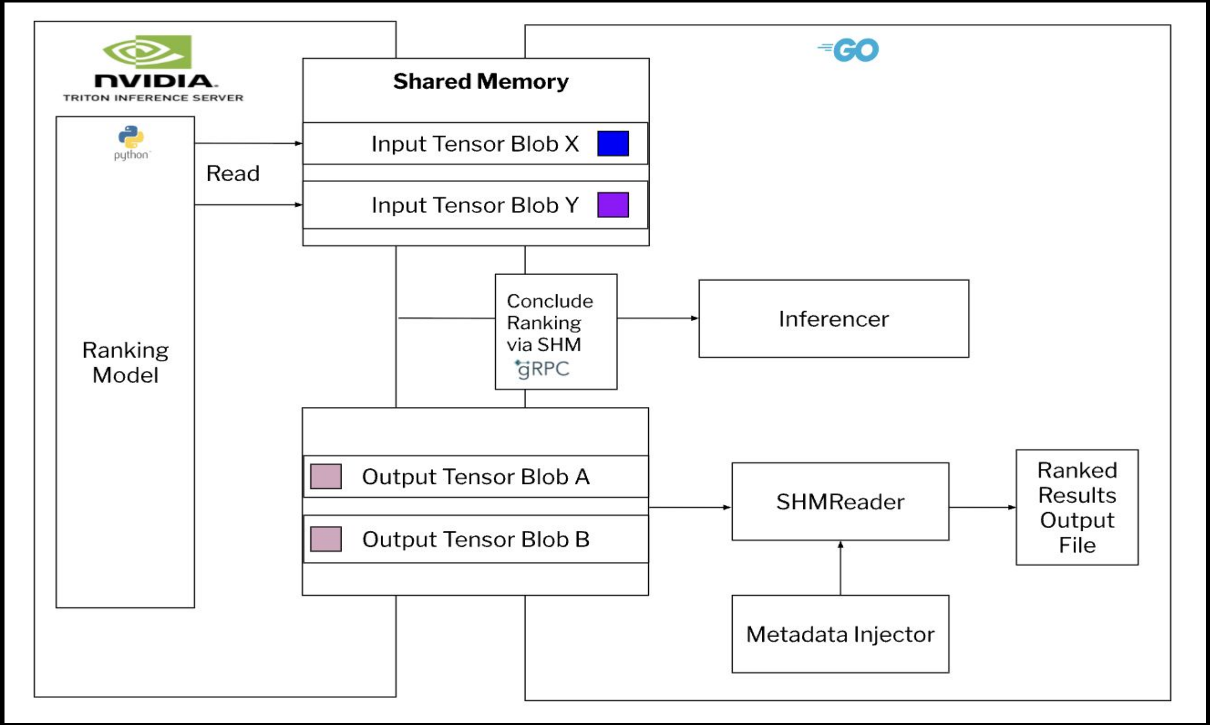
How it all comes together...

Step 6: Output Ranked Results to SHM



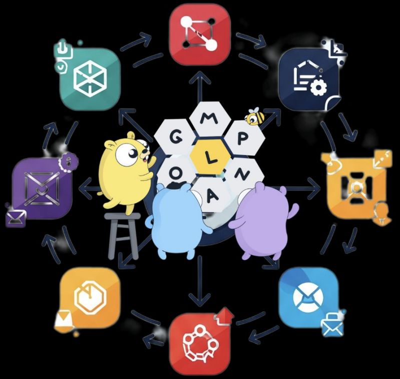
How it all comes together...

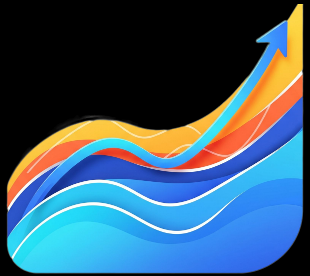
Step 7: Read ranked output from SHMs



Impact

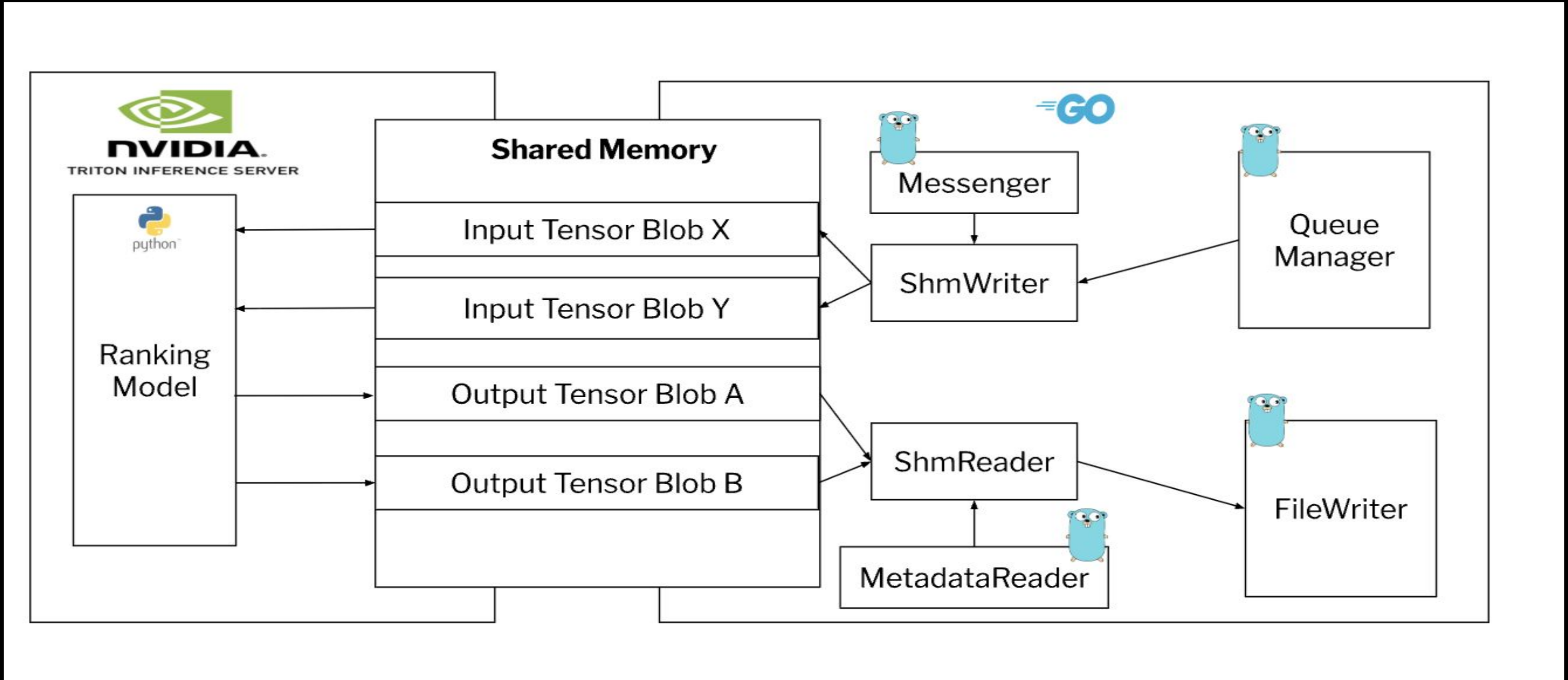
Expand the platform offerings to perform large scale ranking operations using **Go** based tooling to execute ML algorithms written in Python.





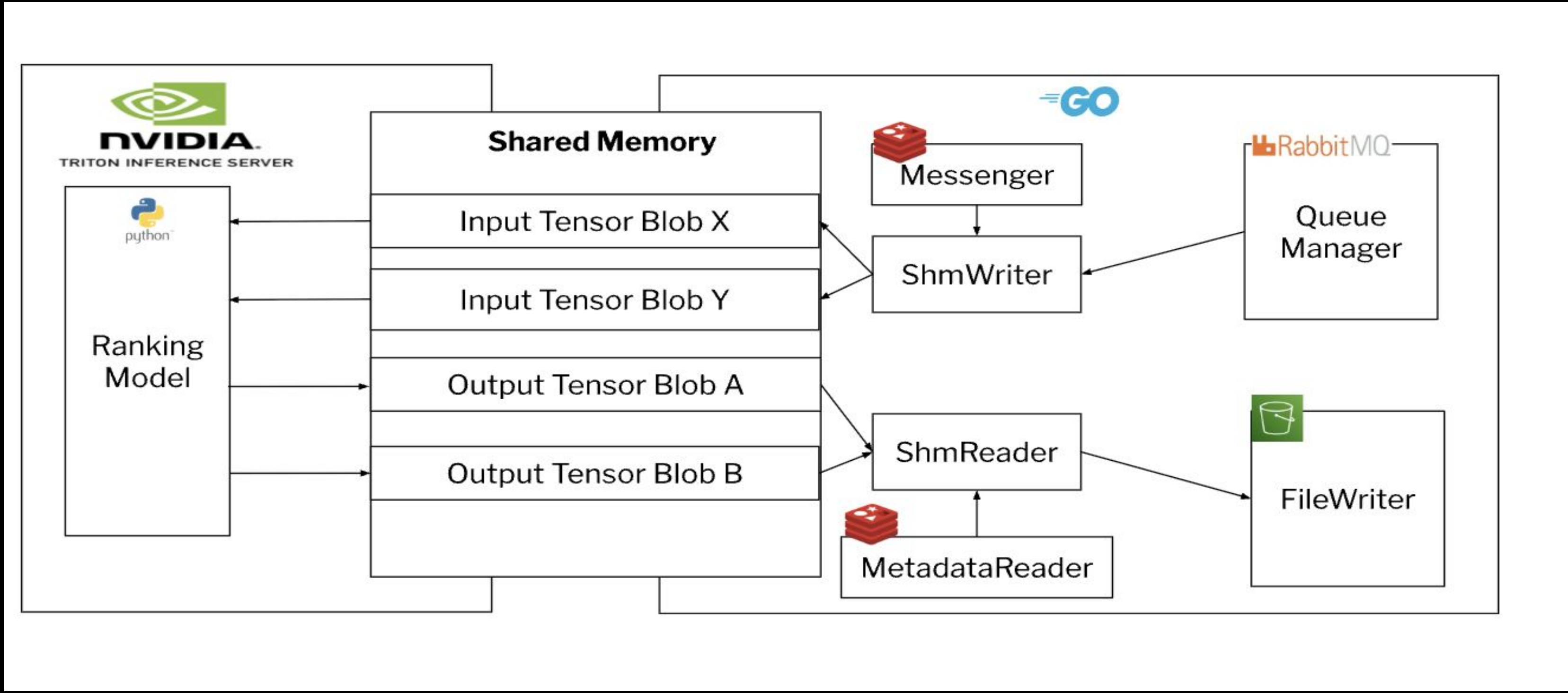
Interfaces in Go

For rapid prototyping and testing and mapping the flow



Interfaces in Go

Swapping out with production clients to different services

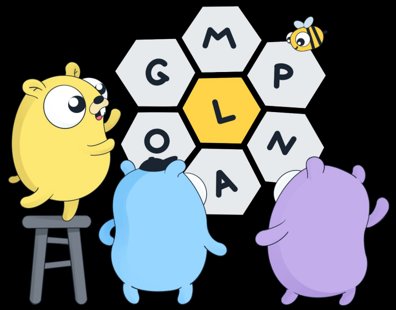


Impact: Targeted Emails

- ⬆ WAU
- ⬆ Weekly clicks
- ⬇ Reduced weekly user-initiated email unsubscribers

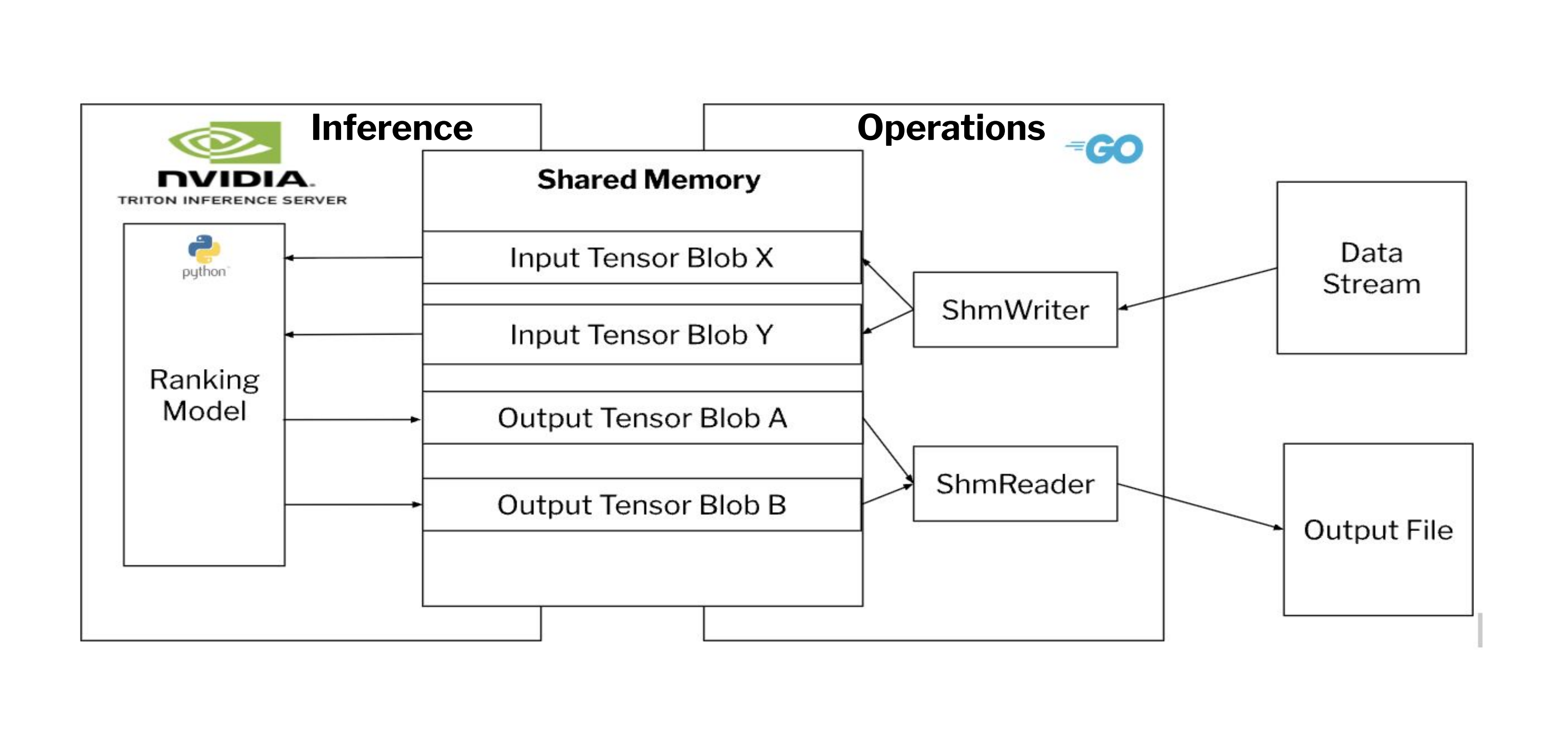
Where do we go from here

Launching a multi-variant version of this service where we **concurrently** perform all of the operations across multiple ranking models.



Questions?

Ranker Service: A Summary



Shm Creation : Read from SHM

Operationalization

```
func readFromSHM(name string) (*Data,error) {

    // Open existing shared memory
    fd, err := syscall.Open(name, syscall.O_RDONLY, 0666)
    if err != nil {
        return nil, fmt.Errorf("failed to open shared memory: %w", err)
    }

    defer syscall.Close(fd)
    // Get file info to determine size
    info, err := syscall.Fstat(fd)
    if err != nil {
        return nil, fmt.Errorf("failed to get shared memory info: %w", err)
    }

    // Map the shared memory
    data, err := syscall.Mmap(fd, 0, int(info.Size),
        syscall.PROT_READ, syscall.MAP_SHARED)
    if err != nil {
        return nil, fmt.Errorf("failed to map shared memory: %w", err)
    }
    defer syscall.Munmap(data)

    // Create a new Data struct
    result := &Data{}

    // Read the data into the struct
    // Assuming the memory layout matches the struct
    offset := 0

    // Read ID (int64)
    result.ID = int64(binary.LittleEndian.Uint64(data[offset:]))
    offset += 8

    // Read Value (float64)



    // Read Length (int32)
    result.Length = int32(binary.LittleEndian.Uint32(data[offset:]))
    offset += 4

    // Read Content
    copy(result.Content[:], data[offset:])

    return result, nil
}
```

The New York Times